

A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

PANCREATITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

5-ASA 5-Aminosalicylic Acid

AKI Acute Kidney Injury

APACHE Acute Physiologic Assessment and Chronic Health Evaluation

CT Computed Tomography

ERCP Endoscopic Retrograde Cholangiopancreatography

GI Gastrointestinal

H Hours

H2RA Histamine-2 Receptor Antagonist

IV Intravenous

MRI Magnetic Resonance Imaging

NSAIDs Non-Steroidal Anti-inflammatory Drugs

PPI Proton Pump Inhibitor

PRN Pro Re Nata (As Needed)

Q Quaque (Every)

QAM Quaque Ante Meridiem (Every Morning)

SSRI Selective Serotonin Reuptake Inhibitor

SNRI Serotonin and Norepinephrine Reuptake Inhibitor

TCA Tricyclic Antidepressants

DEFINITION

Pancreatitis is **inflammation of the pancreas**

It can be mild and self-limiting, but it may also lead to complications that can be life-threatening

Acute Pancreatitis

- **Sudden inflammation** - resolves with appropriate management, but can become severe and cause complications if not properly treated
- Lasts for a few days
- **Can lead to chronic pancreatitis**

Chronic Pancreatitis

- **Ongoing inflammation** usually resulting from multiple bouts of acute pancreatitis
- Resultant damage to the pancreas characteristic of **fibrosis, calcification, and ductal inflammation**
- **Can last for several years**
- **Can predispose patient to pancreatic cancer**

PATHOPHYSIOLOGY

Acute Pancreatitis



Acute injury



Premature activation of
proteolytic enzymes



Autodigestion of
the pancreas

Resulting in inflammation and vascular damage
Long-term inflammation and multiple recurrences →
Chronic pancreatitis

MILD

- No organ failure, local or systemic complications

MODERATE

- Organ failure that resolves within 48 hours and/or
- Local or systemic complications without persistent organ failure

SEVERE

- Persistent organ failure greater than 48 hours
- May involve necrosis

CLINICAL PRESENTATION

Acute Pancreatitis

- Sudden onset of upper abdominal pain that often radiates to the back
- Upper abdominal tenderness
- Acutely ill presentation (e.g. tachycardia, fever)
- Ileus (loss of intestinal function)

- Abdominal distention
- Nausea & vomiting
- Worsening symptoms when lying flat, improvements when leaning forward
- Severe cases: organ failure or mortality

Chronic Pancreatitis

- Constant or intermittent abdominal pain or no pain
- Pancreatic insufficiency (e.g. flatulence, steatorrhea)
- Malabsorption (e.g. unintentional weight loss, fatigue)
- Glucose abnormalities (e.g. intolerance, hypoglycemia)

RISK FACTORS AND COMPLICATIONS

Acute Pancreatitis ➔ Chronic Pancreatitis		
Risk Factors	• Structural causes (e.g. gallstones) • Alcohol consumption • Medications (e.g. azathioprine, 6-mercaptopurine, sulfa drugs, valproate) • Infections (e.g. hepatitis A, mumps, cytomegalovirus) • Metabolic conditions (e.g. obesity, hypertriglyceridemia, hypercalcemia) • Other (e.g. cigarette smoking, pregnancy) • Genetics	• Cigarette smoking • Alcohol consumption • Recurrent acute pancreatitis • Genetics • Idiopathic
Complications	• Organ failure (acute and/or persistent) • Necrosis, infected necrosis, sepsis, shock • Others: Pancreatic abscess, splenic vein thrombosis, glucose abnormalities, bile duct or duodenal obstruction, mortality	

DIAGNOSIS

Acute Pancreatitis

2 of the 3 are required for diagnosis:

- ❑ Abdominal pain
 - ❑ Serum lipase (or amylase) ≥ 3 upper limit of normal
 - ❑ Characteristic findings via CT or MRI
- The APACHE II score is used as a prognostic tool to assess the severity of acute pancreatitis

Imaging:

- Ultrasonography to detect stone obstruction
- CT if severe abdominal pain or suspected complications (e.g. signs of shock, peritonitis)

Chronic Pancreatitis

Clinical history:

- Assessing risk factors and symptoms
- Serum lipase may be normal or elevated

Imaging:

- CT or MRI - first line for diagnosis

GOALS OF THERAPY

- ✓ Address risk factors (e.g. alcohol use, cigarette smoking)
- ✓ Manage symptoms during acute pancreatitis
- ✓ Prevent progression to chronic pancreatitis
- ✓ Prevent complications of pancreatitis and mortality

ACUTE PANCREATITIS MANAGEMENT

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NON-PHARMACOLOGICAL MEASURES

Supportive care:

- **Fluid resuscitation** (e.g. isotonic IV fluids like Ringer's Lactate solution or vasopressor if not fluid responsive)
- Risk factor management (e.g. **smoking and alcohol cessation**, obesity)
- Proper mobilization
- Adequate nutrition:
 - **Mild** acute pancreatitis → regular diet is allowed (low-fat diet is preferred over clear fluids)
 - **Severe** acute pancreatitis → **enteral** feeding within 48 hours significantly reduces pain, medication usage and food intolerance
 - Probiotics → **not recommended**, shown no benefit

Medical Procedures:

- ERCP within 48 hours to remove biliary tract stones

PHARMACOLOGICAL MEASURES

Symptom management is the focus of acute pancreatitis:

- Prophylactic antibiotics are **not recommended**
- Antiemetics can be used to relieve nausea and vomiting
- **Mild pain** → Multimodal analgesic regimen (e.g. NSAIDs, acetaminophen and/or opioids)
- **Moderate – severe pain** → parenteral opioids (e.g. hydromorphone or fentanyl)

Infected Necrosis

- Symptoms: fever, bacteremia, persistent unwellness
- If suspected → Initiate IV empiric antibiotics with good pancreatic penetration to cover pathogens most likely to cause infection

Pathogens	Poor pancreatic penetration	Moderate/good pancreatic penetration
Gram positive: <i>Staphylococcus aureus</i> , <i>epidermidis</i> , <i>faecalis</i> Gram negative: <i>Bacteroides species</i> , <i>Enterobacter species</i> , <i>Escherichia coli</i> , <i>Klebsiella species</i>	Aminoglycosides	Metronidazole Fluoroquinolones Carbapenems Cephalosporins (high dose) Piperacillin/Tazobactam

CHRONIC PANCREATITIS MANAGEMENT

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CHRONIC PANCREATITIS MANAGEMENT

Non-Pharmacological Measures

- Lifestyle and dietary modifications
 - Alcohol & smoking cessation
 - Dietitian referral, smaller and frequent meals
 - Correct fluids and electrolytes if needed
- Surgical intervention in patients with obstruction

Pharmacological Measures

Pain management - stepwise approach:

1. **NSAIDs and/or acetaminophen**
 2. Opioids* - start with lower potency opioids (e.g. tramadol)
 3. Add adjunctive therapy in patients requiring opioids for pain (e.g. TCA, SSRI, SNRI, or gabapentinoids)
- Limited evidence on pancreatic enzyme therapy for pain reduction

*Try to minimize chronic opioid use for pain

MONITORING PARAMETERS

Efficacy Endpoints

Parameter	Degree of Change	Timeframe
Pain and inflammation	Improved and/or resolved	Days to weeks

Safety Endpoints

Parameter	Degree of Change	Timeframe
Prevent and manage complications of acute pancreatitis	Prevent or minimize risk, treat complications	Duration of acute episode
Reduce risk of chronic pancreatitis	Prevent development of chronic pain and loss of pancreatic function	Ongoing

TAKEAWAY

Acute Pancreatitis

- Common causes: gallstones and alcohol
- Non-pharmacological: fluid resuscitation, risk factor management, proper mobilization, adequate nutrition
- Pharmacological: symptom management via analgesics, and anti-emetics if needed
- Infected necrosis – initiate IV empiric therapy with good pancreatic penetration

Chronic Pancreatitis

- Common causes: recurrent acute pancreatitis episodes, smoking, and alcohol use
- Non-pharmacological: lifestyle and dietary modifications
- Pharmacological: Pain management, avoid chronic opioid therapy

CASE

AP is a 33-year-old female presenting to the emergency room with severe epigastric pain (9/10) radiating to her back. She describes the pain as constant, deep, sharp, and worsening after eating and lying down. Her symptoms are also associated with nausea and vomiting. Her past medical history includes hypertension, migraine headaches, anxiety, and hypertriglyceridemia. She currently takes citalopram 10 mg PO QAM, acetaminophen 1000 mg PO Q6H PRN, sumatriptan 25 mg PO PRN, amlodipine 5 mg PO QAM, and atorvastatin 10 mg PO QAM. She binge drinks 6 to 7 beers every Sunday and has one pack of cigarettes each day for the last 10 years. Her diet consists primarily of fast food and she leads a sedentary lifestyle because her job at the restaurant involves long hours of sitting.

1. AP is suspected to have acute pancreatitis. Based on her history and medication profile, which of the following is NOT a risk factor?
 - a) Alcohol consumption
 - b) Cigarette smoke
 - c) Pregnancy
 - d) Hypertriglyceridemia



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AP's labs (lipase is 500 units/L (reference: 0-160 units/L)) and imaging confirm the diagnosis for acute pancreatitis.

2. Which of the following is the most important initial supportive management for AP?
- a) Schedule ERCP within 24 hours and start a no-fat diet
 - b) Initiate IV fluid resuscitation and analgesia
 - c) Initiate IV fluid resuscitation and proper mobilization
 - d) Schedule ERCP within 24 hours and start probiotics



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AP's condition rapidly worsens. She develops a high fever (39.2°C), leukocytosis and expresses feeling persistently unwell. A CT scan confirms pancreatic necrosis and an active infection.

3. Which of the following is NOT appropriate to initiate?
- a) Piperacillin/ Tazobactam
 - b) Ertapenem
 - c) Gentamicin
 - d) Metronidazole



CASE

AP's acute episode resolves, but over the next few years, she has had multiple recurrent attacks. She now reports to the primary care office with concerns of foul-smelling, greasy stools and intermittent abdominal pain (4/10). She also mentioned that she unexpectedly lost 6 kg of weight.

4. Which of the following is the most appropriate for AP's chronic pain?
- a) Gabapentin 400 mg PO once daily
 - b) Venlafaxine XR 75 mg PO daily
 - c) Naproxen 500 mg PO twice daily PRN
 - d) Acetaminophen 325mg/ tramadol 37.5mg PO four times daily



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CHANGE LOG

August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slide 11: Removed mention of somatostatin
- Slide 21: Updated references

November 2025

- Formatting changes throughout, legend updated
- Slide 4: Updated definition of acute pancreatitis
- Slide 6: Updated clinical presentation
- Slide 7: Updated risk factors
- Slide 8: Added new slide on diagnosis
- Slide 9: Added one additional goal of therapy
- Slide 11: Added details to non-pharmacological measures
- Slide 12: Updated pharmacological measures
- Slide 14: Combined pharms and non-pharms
- Slide 16: Updated takeaway
- Slides 17-20: New case and questions added

March 2026

- Slide 2: Updated NAPRA competencies